

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons) — Semester VI Examination, 2013-14

Paper No. MTB-602: Linear Algebra

Time: 3 Hours

Full Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory — 2 marks each.)

- (a) For a vector space V over a field F and $x, y \in V$, $a \in F$, show that:
- (i) $a(x - y) = ax - ay$
 - (ii) $ax = 0 \implies a = 0$ or $x = 0$
- (b) Show that every superset of a linearly dependent set of vectors is linearly dependent.
- (c) Show that the vectors $(1, 2, 1)$, $(2, 1, 0)$, and $(1, -1, 2)$ form a basis of the vector space $\mathbb{R}^3$ over $\mathbb{R}$.
- (d) Let T be a linear transformation on a vector space V such that $T^2 - T + I = 0$. Show that T is invertible.
- (e) Show that similar matrices have the same determinant.
- (f) If v is a characteristic vector of a linear operator T , show that v cannot correspond to more than one characteristic root of T .
- (g) If x and y are vectors in a unitary space V , then prove that $(x, y) = \operatorname{Re}(x, y) + i\operatorname{Re}(x, iy)$.

Question 2.

- (a) Show that a non-empty subset W of a vector space V is a subspace of V if and only if for all $x, y \in W$ and $a \in F$, $ax + y \in W$. Hence determine whether the set of ordered triads (x, y, z) of real numbers with $z > 0$ is a subspace of $\mathbb{R}^3(\mathbb{R})$ or not. [7]
- (b) Show that every linearly independent subset of a finitely generated vector space $V(F)$ is either a basis of V or can be extended to form a basis of V . [7]

Question 3.

- (a) If S and T are subsets of a vector space $V(F)$, then prove that:
- (i) $L(S \cup T) = L(S) + L(T)$
 - (ii) S is a subspace of V if and only if $L(S) = S$.

[7]

- (b) State and prove the rank-nullity theorem for linear transformations. [7]

Question 4.

- (a) Let $T : V \rightarrow V$ be a linear transformation. Prove that the following conditions are equivalent:

(i) $R(T) \cap N(T) = \{0\}$

(ii) $T(T(x)) = 0 \implies T(x) = 0$

[6]

- (b) Find the matrix of the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by:

$$T(a, b, c) = (2b + c, a - 4b, 3a)$$

with respect to the ordered basis $B = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$. [8]

Question 5.

- (a) Let U and V be finite dimensional vector spaces over a field F of dimensions n and m respectively. Prove that the vector space $L(U, V)$ of all linear transformations from U to V is finite dimensional and has dimension mn . [7]

- (b) Let V be an n -dimensional vector space over a field F and let $B = \{x_1, x_2, \dots, x_n\}$ be an ordered basis for V . If $\{a_1, a_2, \dots, a_n\}$ is any ordered set of n scalars from F , show that there exists a unique linear functional f on V such that $f(x_i) = a_i$ for $i = 1, 2, \dots, n$. [7]

Question 6.

- (a) Diagonalize the following matrix:

$$A = \begin{pmatrix} 9 & -4 & -4 \\ -8 & 3 & 4 \\ -16 & 8 & 7 \end{pmatrix}$$

[8]

- (b) Define an inner product space. Let $x = (x_1, x_2, \dots, x_n)$ and $y = (y_1, y_2, \dots, y_n)$ in $\mathbb{C}^n$. Define $(x, y) = x_1\bar{y}_1 + x_2\bar{y}_2 + \dots + x_n\bar{y}_n$. Show that $\mathbb{C}^n$ is an inner product space. [6]

Question 7.

- (a) State and prove the Cauchy-Schwarz inequality. Find when this inequality becomes an equality and supply a proof. [7]
- (b) Define an orthonormal set in an inner product space. Show that:
- (i) Every finite-dimensional inner product space $V \neq \{0\}$ possesses an orthonormal basis.
 - (ii) If $S = \{x_1, x_2, \dots, x_n\}$ is an orthonormal set of vectors in an inner product space V and if a vector y is in the linear span of S , then:

$$y = \sum_{k=1}^n (y, x_k) x_k$$

[7]

OR

Question 7 [Alternate]. Apply the Gram-Schmidt process to the vectors $(1, 0, 1)$, $(1, 3, 1)$, and $(3, 2, 1)$ in $\mathbb{R}^3$ to obtain an orthonormal basis of the inner product space $\mathbb{R}^3$ over $\mathbb{R}$ with respect to the standard inner product. [7]